

(1)

Eulerian and Hamiltonian Graphs:

Lemma 1: if G is a graph in which the degree of each vertex is at least two, then G contains a cycle.

proof 1: if G is a simple graph, let v be any vertex of G . We construct a walk $v \rightarrow v_1 \rightarrow \dots$ inductively by choosing v_1 to be any vertex adjacent to v , the existence of this vertex is guaranteed by our hypothesis, since G has only finite many vertices, we must eventually choose a vertex that has been chosen before.

proof 2: G is not a tree.

Theorem 2 (Euler, 1736):

a connected graph is Eulerian $\iff$ the degree of each vertex is even.

$\Rightarrow$: suppose that P is an Eulerian trail in G , whenever P passes from vertex, there is contribution of 2 towards the degree of that vertex ($\rightarrow v \rightarrow$). Since each edge occurs exactly once in P , each vertex must have even degree.

$\Leftarrow$: the proof is by induction on the number of edges of G . Suppose that the degree of each vertex is even. Since G is connected, each vertex has degree at least 2 and so by (lemma 1) contains a cycle C . If C contains every edge of G the proof is complete, if not we remove from G the edges to form a possibly non disconnected graph H with fewer edges than G .

in which each vertex still has even degree. By induction, each component of H has an Eulerian trail, since each component of H has ~~an Eulerian~~ at least one vertex in common with C , by connectedness, we obtain the required Eulerian ~~graph~~ trail by following the edges of C until a non-isolated vertex of H is reached, tracing the Eulerian trail of component H that contains that vertex, and then continuing along the edges of C ... the whole process terminates when we reach the initial vertex.

~~For~~ this proof can be easily modified to prove the following results

~~Corollary~~ Corollary 1:

a connected graph is Eulerian $\iff$ it can be split (set of edges) into disjoint cycles.

Corollary 2:

~~An Eulerian Graph can be~~

a connected graph is semi-Eulerian $\iff$ exactly two vertices of odd degree.

$\Rightarrow$ Assume we have trail T from u to v , then ~~add~~ if we add an edge between v and u we obtain Eulerian Graph (all vertices have even degree), if we remove that edge only u and v have odd degree.

$\Leftarrow$ we add edge between this two vertices, then all vertices have even degree then it is Eulerian graph.

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Theorem 3 (Fleury's algorithm):

Let G be an Eulerian graph, then the following construction is always possible and produces an Eulerian trail of G . Start at any vertex u and traverse the edges in any way subject to this rule;

1. erase the edge as they are traversed and if any isolated edge result, erase them too.
2. at each stage, use a bridge only if there ~~exists~~ is no alternative.

proof: we first show that the construction can be carried out at each stage, suppose that we have just reached a vertex u , $y \neq u$

then the subgraph H that remains is connected and contains only two vertices of odd degree

u and v . to show that the construction can

be carried out, we must show that the removal

of the next edge doesn't disconnect H or equivalently,

that v is incident with at most one bridge. But if this is not the case,

then there exists a bridge vw such that the component K of $H - vw$ containing w doesn't contain u , since the vertex w has odd degree in K , some other vertex of K must also have odd degree, giving the required vertex v . $y \neq u$

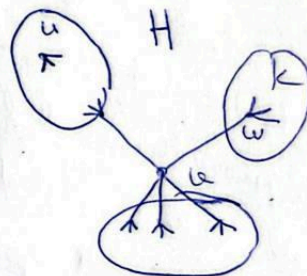
the proof is almost identical, as long as there are still edges incident with u .

it remains only to show that this construction always yields a complete Eulerian

trail, but this is clear, since there are no edges of G ~~containing~~ remaining untraversed

when the last edge incident to u is used, since otherwise the removal of some

other edge adjacent to one of these edges would have disconnected the graph.



Contradicting rule 2.

Theorem 4;

a strongly connected digraph is Eulerian $\iff$ for each vertex v of D $\text{outdeg}(v) = \text{indeg}(v)$.

proof;

$\Rightarrow$ suppose D has an eulerian trail, each time the trail enters a vertex v it uses one incoming arc, each time it leaves v , it uses one outgoing arc, since every arc appears exactly once in P , then $\text{outdeg}(v) = \text{indeg}(v)$.

$\Leftarrow$ we proceed by induction on the number of arcs, the result is trivial for no arcs.

assume theorem holds for all connected digraphs with fewer arcs than D .

since D is connected and every vertex satisfies $\text{outdeg}(v) = \text{indeg}(v)$,

~~every vertex has at least one outgoing arc and repeatedly following outgoing arcs~~

every vertex has at least one outgoing arc. starting from any vertex and repeatedly following outgoing arcs, because the graph is finite,

we must eventually repeat a vertex. Hence D contains a directed cycle C .

if C ~~contains~~ contain all arcs, the proof is complete, else we remove

arcs ~~from~~ of C from D , obtaining a digraph possibly disconnected H ,

each component of H has fewer arcs than D then it has an eulerian trail.

moreover, since D is connected, every component of H shares at least

one vertex with the cycle C . we now construct eulerian trail of D .

start traversing the directed cycle C , when a vertex is reached that belongs to H with unused arcs, traverse eulerian trail of H , continue along C ...

the resulting is directed Eulerian trail of D .